Poster: A Fistful of Queries: Accurate and Lightweight Public DNS Anycast Enumeration

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Abstract

In this study, we present PublicDNSEnum, a lightweight and accurate tool to enumerate anycast instances of public DNS services. Leveraging the open and forwarding features of DNS resolvers, PublicDNSEnum enables researchers to trigger and conduct anycast enumeration of public DNS anycast instances on a single vantage host. It transforms a massive number of open forwarders into distributed measurement nodes by utilizing the forwarding relationship between open forwarders and public DNS. Experiments on Google Public DNS show that PublicDNSEnum achieves better recall than the state-of-the-art technique, including iGreedy.

CCS Concepts

• **Security and privacy → Security protocols**.

Keywords

Anycast enumeration, Iterative algorithm, Open forwarders, Forwarding relationship

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1 Introduction

Anycast is a technique that allows service providers to offer their services via a uniform service IP address while deploying service instances in geographically distributed places. Public DNS service providers favor anycast to improve the domain name system's resistance to attacks and load-balancing performance. Anycast enumeration is often used to facilitate anycast systems' deployment, performance, quality of service, and resistance.

In this study, we present PublicDNSEnum, a lightweight, accurate tool to enumerate anycast instances of public DNS services. PublicDNSEnum is inspired by the observation that a large number of open forwarders in the wild Internet forward DNS requests to public DNS services. PublicDNSEnum leverages such open forwarders into distributed vantage points by sending elaborately designed domain queries and domain resolution logic so that PublicDNSEnum enables researchers to trigger and conduct anycast enumeration of public DNS anycast instances on a single vantage host.

2 Methodology

Client-side DNS Infrastructure. Client-side DNS infrastructure can be roughly categorized into three components: open resolvers (ODNSes), hidden resolvers (HDNSes), and recursive resolvers (RDNSes) [3]. As illustrated in Figure 1, the direction of the arrow represents the direction of the DNS traffic flow. Furthermore, public DNS services generally composed of two or several service IPs, multiple HDNSes and RDNS$_i$es. RDNS$_i$es act as the egress points or the replicas of anycast system.

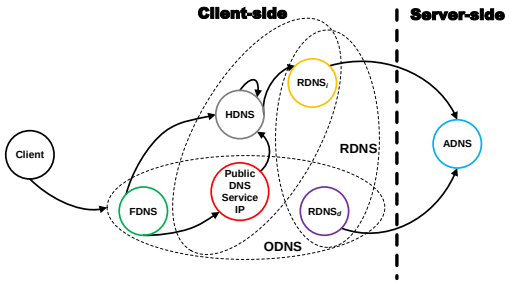

Figure 1: Client-side DNS Infrastructure

Problem Statement. The problem we are trying to solve is: given anycast-based public DNS service IP addresses, enumerate the replicas and their IPs behind the service IPs, or equivalently expressed as discover the resolver pools behind given public DNS service.

Workflow. Figure 2 shows the workflow from a high-level point of view. We first discover the forwarding relationships among resolvers as initial data using an Internet-wide scan and detect the forwarding relationships of the given public DNS service IPs. Next,

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we iteratively perform forwarding relationship measurement, aggregation of RDNS$_i$es, and correlation of FDNSes by feeding the open forwarders to PublicDNSEnum at step k back to update the selection of vantage points at step k+1.

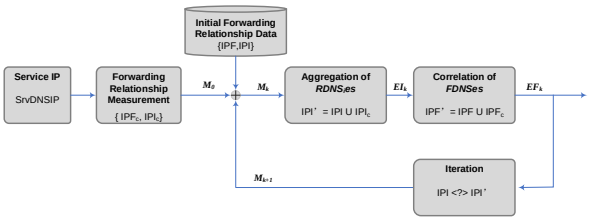

Figure 2: Methodology Workflow

3 Validation

We validate PublicDNSEnum on publicly available ground truth and compare our methodology against state-of-art methods, i.e., iGreedy[1], CNAME-chain-based[2], and NS-chain-based methods [4] in terms of recall.

Ground Truth. Given the prevalence of anycast-based public DNS services, few public DNS service providers make their replicas public, except Google, whose service IP addresses are 8.8.8.8 and 8.8.4.4 who publishes IP address ranges of their anycast service on their websites, consisting of 292 IPv4 prefixes ranging from /24 to /26. Google aggregates these IP ranges into 48 unique Geo-associated codes.

Preliminary Results. Recall of anycast replicas is a core metric in anycast enumeration, which be calculated as follows:

$$Recall = \frac{sum_{dis}}{sum_{total}} \times 100\%$$

As Google only publishes the IP ranges of its anycast replicas, the exact instances of Google Public DNS remain unknown to us. However, we performed Ping tests (ICMP echo) on all the IPs in the IP ranges published by Google and found that 47 of 292 IP ranges don't even have one host alive, which implies these IP ranges are inactive. In order to quantify the performance of PublicDNSEnum and iGreedy on different granularity, we defined three recall metrics: $Recall_{ip}$, $Recall_{ipprefix}$, $Recall_{ipgeocode}$, which represents instance *Recall* at IP level, IP prefix level, and IP geocode level, respectively.

Table 1: Test Results of Google Public DNS

Method	$Recall_{ip}$	$Recall_{ipprefix}$	$Recall_{ipgeocode}$
iGreedy[1]	N/A	N/A	39.58%
CNAME Chain [2]	4.50%	N/A	N/A
NS Chain[4]	7.19%	29.79%	45.83%
PublicDNSEnum	**33.29%**	**58.22%**	**62.50%**

As Table 1 shows, iGreedy used a latency-based enumeration technique, so $Recall_{ip}$ and $Recall_{ipprefix}$ didn't apply for it. iGreedy can only calculate $Recall_{ipgeocode}$. iGreedy enumerated 19 unique Geo-associated codes out of 48 in total, yielding a $Recall_{ipgeocode}$ of 39.58%, while PublicDNSEnum enumerated 30 out of 48 Geocodes, yielding a $Recall_{ipgeocode}$ of 62.50%.

4 Conclusion

This study presents PublicDNSEnum, a lightweight, accurate tool to enumerate anycast instances of public DNS services. Compared to traditional methods, PublicDNSEnum enables researchers to launch anycast enumeration of public DNS from a single vantage host. Experiments on Google Public DNS show that the $Recall_{ipgeocode}$ of PublicDNSEnum is 1.58 times than that of iGreedy. Thanks to its low-cost feature, PublicDNSEnum is expected to promote lightweight anycast service evaluation and optimization research.

Acknowledgments

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